

## E-Commerce Sales Analysis for Data-Driven Decision Making

### Task 1 : Subtask 1: Research Indian E-Commerce Market

#### Research on Indian E-Commerce Market and Data-Driven Strategies

##### Introduction

The Indian e-commerce market has grown rapidly over the past decade due to increasing internet penetration, smartphone usage, digital payments, and changing consumer behavior. Customers now prefer online shopping because it offers convenience, variety, competitive pricing, and home delivery services.

India is currently one of the fastest-growing e-commerce markets in the world. Major platforms such as Flipkart, Amazon India, Meesho, Nykaa, Myntra, and Ajio dominate the Indian online retail sector. These companies heavily rely on data analytics to improve customer satisfaction, increase revenue, optimize logistics, and make better business decisions

##### Major Indian E-Commerce Players and Their Business Models

| Company      | Business Model    | Main Products/Services                |
|--------------|-------------------|---------------------------------------|
| Flipkart     | Marketplace (B2C) | Electronics, fashion, appliances      |
| Amazon India | Marketplace (B2C) | General merchandise                   |
| Meesho       | Social Commerce   | Affordable products, reseller network |
| Nykaa        | B2C + D2C         | Beauty, skincare, fashion             |
| Myntra       | Marketplace       | Fashion and lifestyle                 |
| Ajio         | D2C + Marketplace | Fashion and accessories               |

##### Flipkart

Flipkart is one of India's largest e-commerce companies and is well known for electronics and smartphone sales. It attracts customers through major sale events such as Big Billion Days.

##### Amazon India

Amazon India operates using a marketplace model where third-party sellers list products. Its fast delivery system through Amazon Prime gives it a competitive advantage.

## **Meesho**

Meesho focuses on low-cost products and targets customers from Tier 2 and Tier 3 cities. It also allows small resellers to earn profits by sharing product listings.

## **Nykaa**

Nykaa mainly sells beauty and cosmetic products and has built strong brand loyalty among women customers.

## **Myntra and Ajio**

Myntra and Ajio dominate online fashion retail by offering personalized recommendations and trendy products.

## **Data-Driven Strategies Used by E-Commerce Companies**

### **1. Personalized Marketing**

E-commerce companies collect customer browsing history, search behavior, and purchase history to recommend products.

Example:

If a customer searches for shoes on Myntra, the platform recommends similar products and sends discount notifications.

Benefits:

- Higher customer engagement
- Increased sales
- Better customer experience

### **2. Dynamic Pricing**

Companies frequently change product prices based on:

- Customer demand
- Competitor prices
- Seasonal trends
- Inventory levels

For example, Amazon India often provides discounts during Great Indian Festival.

### 3. Logistics Optimization

Companies use predictive analytics to improve delivery operations by:

- Forecasting demand
- Managing warehouse inventory
- Reducing delivery time
- Optimizing routes

Flipkart uses strong logistics systems to handle large sales volumes.

### 4. Fraud Detection

Analytics helps companies detect:

- Fake product returns
- Fraud payments
- Fake customer accounts

This reduces business losses.

## Key Metrics Used in E-Commerce Analytics

### Sales Metrics

- Revenue
- Average Order Value (AOV)
- Repeat Purchase Rate

$$AOV = \frac{\text{Total Revenue}}{\text{Total Orders}}$$

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### Customer Metrics

- Customer Retention Rate
- Customer Lifetime Value (CLV)
- Customer Acquisition Cost (CAC)

$$CAC = \frac{\text{Marketing Cost}}{\text{New Customers}}$$

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## Marketing Metrics

- Click Through Rate (CTR)
- Conversion Rate
- Cost Per Click (CPC)

$$\text{Conversion Rate} = \frac{\text{Purchases}}{\text{Visitors}} \times 100$$

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## Operational Metrics

- Order fulfilment time
  - Return rate
  - Inventory turnover
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## Seasonal Sales Impact

Major sale events such as Big Billion Days, Great Indian Festival, and Meesho Mega Blockbuster Sale significantly increase customer purchases.

During these periods companies:

- Increase inventory stock
- Offer discounts
- Spend more on advertisements
- Improve delivery capacity

These strategies help companies maximize profits during festive seasons like Diwali.

## Key Insights / Findings

### 1. Flipkart and Amazon India dominate the Indian e-commerce market

These platforms have large customer bases, strong logistics networks, and major seasonal sale events that help them generate high revenue.

**Insight:** Companies with better delivery systems and wider product variety attract more customers.

## **2. Personalized recommendations increase sales**

Platforms like Myntra and Nykaa use customer browsing history and purchase behavior to recommend relevant products.

**Insight:** Personalized marketing improves customer engagement and increases conversion rates.

## **3. Seasonal sales significantly boost revenue**

Events like Big Billion Days and Great Indian Festival create huge spikes in sales.

**Insight:** Proper inventory planning and marketing strategies are important during festive seasons.

## **4. Data analytics helps improve delivery efficiency**

Companies predict customer demand and store products in warehouses before major sales.

**Insight:** Better planning reduces delivery delays and improves customer satisfaction.

## **5. Customer retention is as important as acquiring new customers**

Companies spend heavily on ads to attract new users, but retaining existing customers helps increase long-term profits.

**Insight:** High customer retention improves Customer Lifetime Value (CLV).

## **6. Return rates can affect profitability**

Fashion platforms like Myntra and Ajio may face higher return rates due to size issues.

**Insight:** Reducing returns can improve profits.

## **7. Data-driven pricing improves competitiveness**

Platforms frequently change prices based on market demand and competitor pricing.

**Insight:** Dynamic pricing helps companies remain competitive and maximize sales.

## **Conclusion**

The Indian e-commerce industry is highly competitive and data-driven. Companies use analytics to understand customer behavior, optimize pricing strategies, improve logistics, and increase profits. Seasonal sales events further highlight the importance of data analysis in managing demand and improving operational efficiency. For future data analysis tasks, understanding these concepts will help in analyzing customer trends, sales performance, and business growth.

## task 2: Download and Explore Dataset

Dataset(superstore.csv)

### Dataset Observations

- The dataset contains sales, customer, product, and shipping-related information.
- Total dataset size: *(mention your actual rows and columns from df.shape)*  
Example: **51290 rows and 27 columns**
- Important columns identified for analysis include:
  - Order.ID
  - Order.Date
  - Customer.ID
  - Product.Name
  - Quantity
  - Sales
  - Profit
  - Region
  - State
- Order.Date and Ship.Date were initially stored as **object datatype** and were converted into proper datetime format.
- Missing values check: "No major missing values were found."
- Duplicate check:  
"No duplicate records were found."
- The dataset is suitable for performing sales analysis, customer segmentation, regional analysis, and product performance analysis

## Subtask 3: Data Cleaning and Preprocessing

### final observations

- Missing values were checked and handled.

- No duplicate records were found.
- Date columns were converted into datetime format.
- Categorical columns were checked for inconsistencies.
- No invalid negative sales/quantity values were found.
- Outliers were identified in Sales, Profit, and Shipping Cost but retained as they represent real business transactions.
- New time-based columns were created for future trend analysis.
- Final cleaned dataset was saved for further analysis.